FlightDB – Airport Management System

Query Outputs

1. **Creating Database**

   CREATE DATABASE FlightDB

2. **Insert Records**

   INSERT INTO Airport SET code = 'MAA', name = 'Chennai International Airport', city = 'Chennai', country = 'India'

3. **Update**

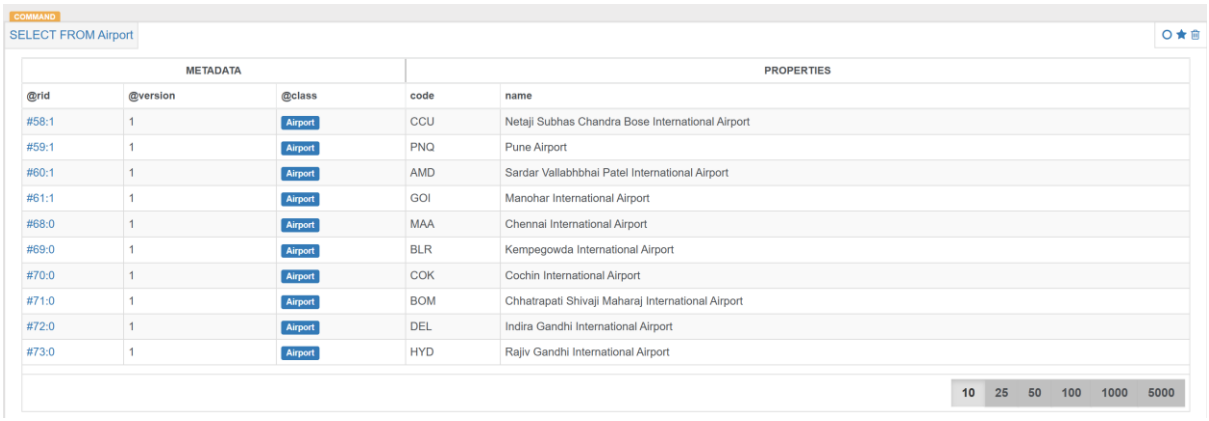
   UPDATE Booking SET status = 'Cancelled' WHERE bookingId = 'BK1001'

4. **Delete**

   DELETE EDGE MakesBooking WHERE out.passengerId = 'P001' AND in.bookingId = 'BK1001'

5. **Display All Airport Records**

   SELECT FROM Airport


COMMAND

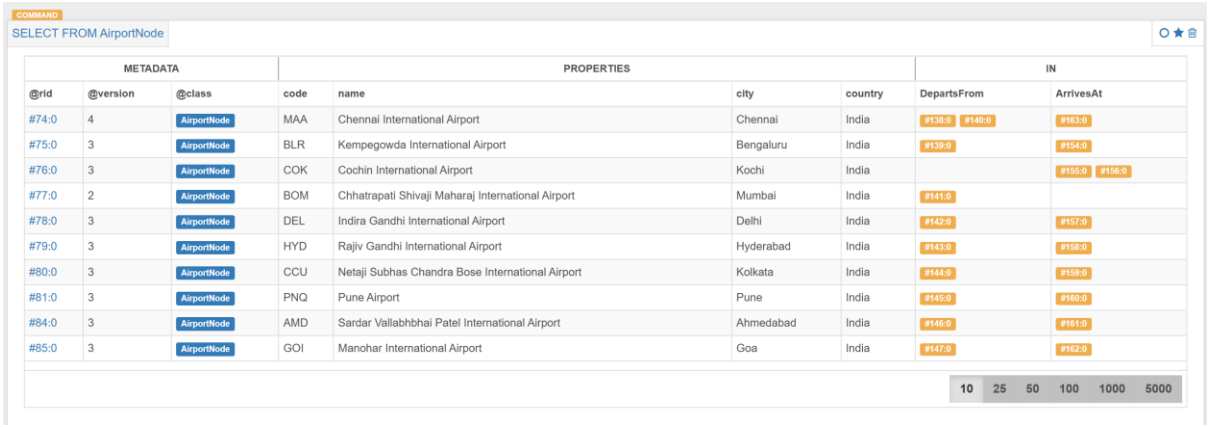
SELECT FROM Airport

METADATA			PROPERTIES	
@rid	@version	@class	code	name
#58:1	1	Airport	CCU	Netaji Subhas Chandra Bose International Airport
#59:1	1	Airport	PNQ	Pune Airport
#60:1	1	Airport	AMD	Sardar Vallabhbhai Patel International Airport
#61:1	1	Airport	GOI	Manohar International Airport
#68:0	1	Airport	MAA	Chennai International Airport
#69:0	1	Airport	BLR	Kempegowda International Airport
#70:0	1	Airport	COK	Cochin International Airport
#71:0	1	Airport	BOM	Chhatrapati Shivaji Maharaj International Airport
#72:0	1	Airport	DEL	Indira Gandhi International Airport
#73:0	1	Airport	HYD	Rajiv Gandhi International Airport

10 25 50 100 1000 5000

6. **Display All Airport Vertices**

SELECT FROM AirportNode



COMMAND

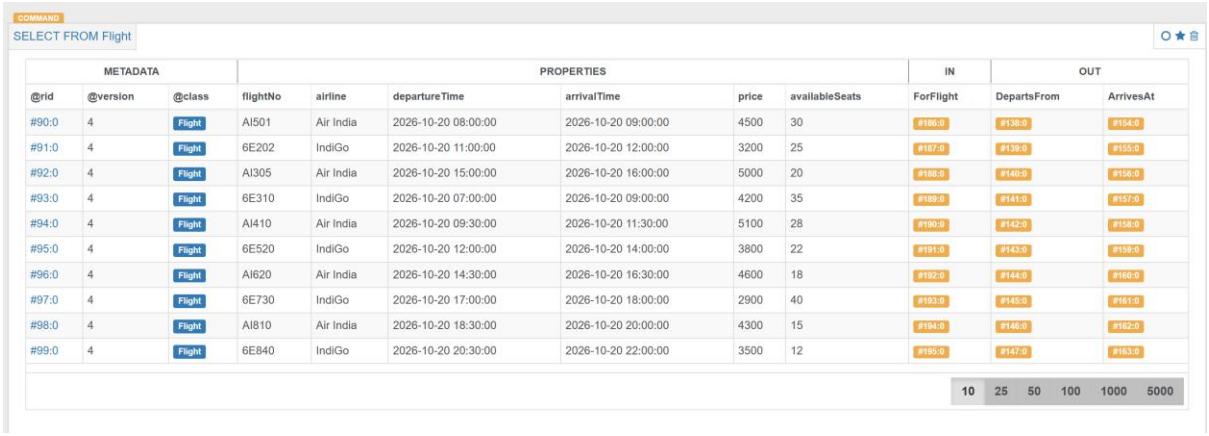
SELECT FROM AirportNode

METADATA			PROPERTIES				IN	
@rid	@version	@class	code	name	city	country	DepartsFrom	ArrivesAt
#74:0	4	AirportNode	MAA	Chennai International Airport	Chennai	India	#138:0 #140:0	#163:0
#75:0	3	AirportNode	BLR	Kempegowda International Airport	Bengaluru	India	#139:0	#154:0
#76:0	3	AirportNode	COK	Cochin International Airport	Kochi	India		#155:0 #156:0
#77:0	2	AirportNode	BOM	Chhatrapati Shivaji Maharaj International Airport	Mumbai	India	#141:0	
#78:0	3	AirportNode	DEL	Indira Gandhi International Airport	Delhi	India	#142:0	#157:0
#79:0	3	AirportNode	HYD	Rajiv Gandhi International Airport	Hyderabad	India	#143:0	#158:0
#80:0	3	AirportNode	CCU	Netaji Subhas Chandra Bose International Airport	Kolkata	India	#144:0	#159:0
#81:0	3	AirportNode	PNQ	Pune Airport	Pune	India	#145:0	#160:0
#84:0	3	AirportNode	AMD	Sardar Vallabhbhai Patel International Airport	Ahmedabad	India	#146:0	#161:0
#85:0	3	AirportNode	GOI	Manohar International Airport	Goa	India	#147:0	#162:0

10 25 50 100 1000 5000

7. **Display All Flight Records**

SELECT FROM Flight



COMMAND

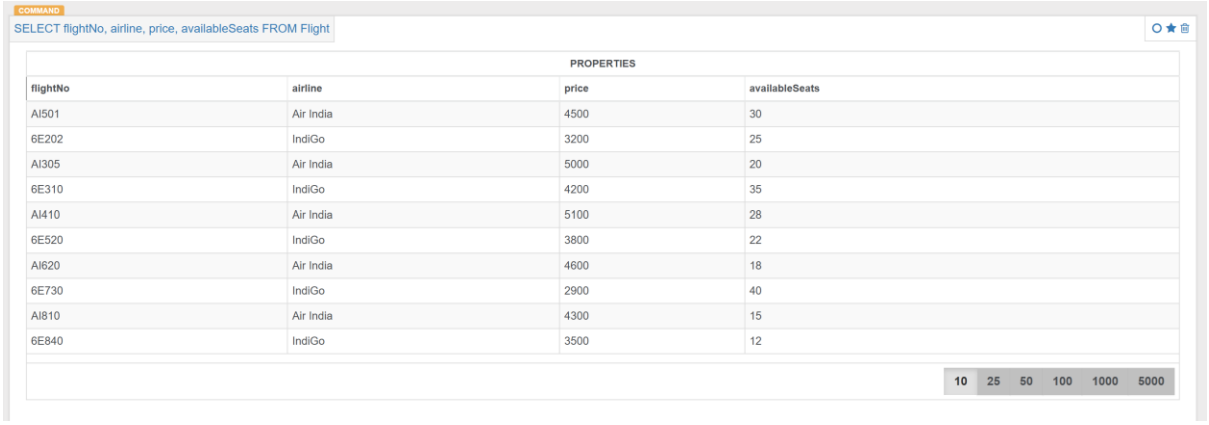
SELECT FROM Flight

METADATA			PROPERTIES						IN	OUT	
@rid	@version	@class	flightNo	airline	departureTime	arrivalTime	price	availableSeats	ForFlight	DepartsFrom	ArrivesAt
#90:0	4	Flight	AI501	Air India	2026-10-20 08:00:00	2026-10-20 09:00:00	4500	30	#186:0	#138:0	#154:0
#91:0	4	Flight	6E202	IndiGo	2026-10-20 11:00:00	2026-10-20 12:00:00	3200	25	#187:0	#139:0	#155:0
#92:0	4	Flight	AI305	Air India	2026-10-20 15:00:00	2026-10-20 16:00:00	5000	20	#188:0	#140:0	#156:0
#93:0	4	Flight	6E310	IndiGo	2026-10-20 07:00:00	2026-10-20 09:00:00	4200	35	#189:0	#141:0	#157:0
#94:0	4	Flight	AI410	Air India	2026-10-20 09:30:00	2026-10-20 11:30:00	5100	28	#190:0	#142:0	#158:0
#95:0	4	Flight	6E520	IndiGo	2026-10-20 12:00:00	2026-10-20 14:00:00	3800	22	#191:0	#143:0	#159:0
#96:0	4	Flight	AI620	Air India	2026-10-20 14:30:00	2026-10-20 16:30:00	4600	18	#192:0	#144:0	#160:0
#97:0	4	Flight	6E730	IndiGo	2026-10-20 17:00:00	2026-10-20 18:00:00	2900	40	#193:0	#145:0	#161:0
#98:0	4	Flight	AI810	Air India	2026-10-20 18:30:00	2026-10-20 20:00:00	4300	15	#194:0	#146:0	#162:0
#99:0	4	Flight	6E840	IndiGo	2026-10-20 20:30:00	2026-10-20 22:00:00	3500	12	#195:0	#147:0	#163:0

10 25 50 100 1000 5000

8. **Display Flight Details**

SELECT flightNo, airline, price, availableSeats FROM Flight



COMMAND

SELECT flightNo, airline, price, availableSeats FROM Flight

PROPERTIES			
flightNo	airline	price	availableSeats
AI501	Air India	4500	30
6E202	IndiGo	3200	25
AI305	Air India	5000	20
6E310	IndiGo	4200	35
AI410	Air India	5100	28
6E520	IndiGo	3800	22
AI620	Air India	4600	18
6E730	IndiGo	2900	40
AI810	Air India	4300	15
6E840	IndiGo	3500	12

10 25 50 100 1000 5000

9. **Display All Passenger Records**

SELECT FROM Passenger

COMMAND

SELECT FROM Passenger

METADATA			PROPERTIES			OUT
@rid	@version	@class	passengerId	name	email	MakesBooking
#106:0	4	Passenger	P001	Ananya Sharma	ananya@example.com	#170:0
#107:0	2	Passenger	P002	Rahul Kumar	rahul@example.com	#171:0
#109:0	2	Passenger	P003	Priya Nair	priya@example.com	#172:0
#110:0	2	Passenger	P004	Arjun Menon	arjun@example.com	#173:0
#111:0	2	Passenger	P005	Meera Iyer	meera@example.com	#174:0
#112:0	2	Passenger	P006	Karthik Rao	karthik@example.com	#175:0
#113:0	2	Passenger	P007	Nisha Patel	nisha@example.com	#176:0
#114:0	2	Passenger	P008	Vikram Das	vikram@example.com	#177:0
#115:0	2	Passenger	P009	Divya Shah	divya@example.com	#178:0
#116:0	2	Passenger	P010	Rohan Verma	rohan@example.com	#179:0

1 2

10 25 50 100 1000 5000

10. Display Passenger Details

SELECT passengerId, name, email FROM Passenger

COMMAND

SELECT passengerId, name, email FROM Passenger

PROPERTIES		
passengerId	name	email
P001	Ananya Sharma	ananya@example.com
P002	Rahul Kumar	rahul@example.com
P003	Priya Nair	priya@example.com
P004	Arjun Menon	arjun@example.com
P005	Meera Iyer	meera@example.com
P006	Karthik Rao	karthik@example.com
P007	Nisha Patel	nisha@example.com
P008	Vikram Das	vikram@example.com
P009	Divya Shah	divya@example.com
P010	Rohan Verma	rohan@example.com

1 2

10 25 50 100 1000 5000

11. Display All Booking Records

SELECT FROM Booking

COMMAND

SELECT FROM Booking

METADATA			PROPERTIES					IN	OUT
@rid	@version	@class	bookingId	bookingDate	seats	totalPrice	status	MakesBooking	ForFlight
#122:0	3	Booking	BK1001	2026-10-13 10:00:00	2	9000	CONFIRMED	#170:0	#186:0
#123:0	3	Booking	BK1002	2026-10-13 10:30:00	1	3200	CONFIRMED	#171:0	#187:0
#124:0	3	Booking	BK1003	2026-10-13 11:00:00	1	5000	CONFIRMED	#172:0	#188:0
#125:0	3	Booking	BK1004	2026-10-13 11:15:00	2	8400	CONFIRMED	#173:0	#189:0
#126:0	3	Booking	BK1005	2026-10-13 11:30:00	1	5100	CONFIRMED	#174:0	#190:0
#127:0	3	Booking	BK1006	2026-10-13 11:45:00	2	7600	CONFIRMED	#175:0	#191:0
#128:0	3	Booking	BK1007	2026-10-13 12:00:00	1	4600	CONFIRMED	#176:0	#192:0
#129:0	3	Booking	BK1008	2026-10-13 12:15:00	3	8700	CONFIRMED	#177:0	#193:0
#130:0	3	Booking	BK1009	2026-10-13 12:30:00	1	4300	CONFIRMED	#178:0	#194:0
#131:0	3	Booking	BK1010	2026-10-13 12:45:00	2	7000	CONFIRMED	#179:0	#195:0

10 25 50 100 1000 5000

12. Display Booking Details

SELECT bookingId, seats, totalPrice, status FROM Booking

COMMAND

SELECT bookingId, seats, totalPrice, status FROM Booking

PROPERTIES			
bookingId	seats	totalPrice	status
BK1001	2	9000	CONFIRMED
BK1002	1	3200	CONFIRMED
BK1003	1	5000	CONFIRMED
BK1004	2	8400	CONFIRMED
BK1005	1	5100	CONFIRMED
BK1006	2	7600	CONFIRMED
BK1007	1	4600	CONFIRMED
BK1008	3	8700	CONFIRMED
BK1009	1	4300	CONFIRMED
BK1010	2	7000	CONFIRMED

10 25 50 100 1000 5000

13. Display Departure Relationships

SELECT FROM DepartsFrom

COMMAND

SELECT FROM DepartsFrom

METADATA			PROPERTIES	
@rid	@version	@class	out	in
#138:0	1	DepartsFrom	#90:0	#74:0
#139:0	1	DepartsFrom	#91:0	#75:0
#140:0	1	DepartsFrom	#92:0	#74:0
#141:0	1	DepartsFrom	#93:0	#77:0
#142:0	1	DepartsFrom	#94:0	#78:0
#143:0	1	DepartsFrom	#95:0	#79:0
#144:0	1	DepartsFrom	#96:0	#80:0
#145:0	1	DepartsFrom	#97:0	#81:0
#146:0	1	DepartsFrom	#98:0	#84:0
#147:0	1	DepartsFrom	#99:0	#85:0

10 25 50 100 1000 5000

14. Display Arrival Relationships

SELECT FROM ArrivesAt

COMMAND

SELECT FROM ArrivesAt

METADATA			PROPERTIES	
@rid	@version	@class	out	in
#154:0	1	ArrivesAt	#90:0	#75:0
#155:0	1	ArrivesAt	#91:0	#76:0
#156:0	1	ArrivesAt	#92:0	#76:0
#157:0	1	ArrivesAt	#93:0	#78:0
#158:0	1	ArrivesAt	#94:0	#79:0
#159:0	1	ArrivesAt	#95:0	#80:0
#160:0	1	ArrivesAt	#96:0	#81:0
#161:0	1	ArrivesAt	#97:0	#84:0
#162:0	1	ArrivesAt	#98:0	#85:0
#163:0	1	ArrivesAt	#99:0	#74:0

10 25 50 100 1000 5000

15. Display Passenger–Booking Relationships

SELECT FROM MakesBooking

COMMAND

SELECT FROM MakesBooking

METADATA			PROPERTIES	
@rid	@version	@class	out	in
#170:0	1	MakesBooking	#106:0	#122:0
#171:0	1	MakesBooking	#107:0	#123:0
#172:0	1	MakesBooking	#109:0	#124:0
#173:0	1	MakesBooking	#110:0	#125:0
#174:0	1	MakesBooking	#111:0	#126:0
#175:0	1	MakesBooking	#112:0	#127:0
#176:0	1	MakesBooking	#113:0	#128:0
#177:0	1	MakesBooking	#114:0	#129:0
#178:0	1	MakesBooking	#115:0	#130:0
#179:0	1	MakesBooking	#116:0	#131:0

10 25 50 100 1000 5000

16. Display Booking–Flight Relationships

SELECT FROM ForFlight

COMMAND

SELECT FROM ForFlight

METADATA			PROPERTIES	
@rid	@version	@class	out	in
#186:0	1	ForFlight	#122:0	#90:0
#187:0	1	ForFlight	#123:0	#91:0
#188:0	1	ForFlight	#124:0	#92:0
#189:0	1	ForFlight	#125:0	#93:0
#190:0	1	ForFlight	#126:0	#94:0
#191:0	1	ForFlight	#127:0	#95:0
#192:0	1	ForFlight	#128:0	#96:0
#193:0	1	ForFlight	#129:0	#97:0
#194:0	1	ForFlight	#130:0	#98:0
#195:0	1	ForFlight	#131:0	#99:0

10 25 50 100 1000 5000

17. Count Airport Vertices

SELECT COUNT(*) AS total FROM AirportNode

COMMAND

SELECT COUNT(*) AS total FROM AirportNode

PROPERTIES
total
10

10 25 50 100 1000 5000

18. Count Flight Records

SELECT COUNT(*) AS total FROM Flight

COMMAND

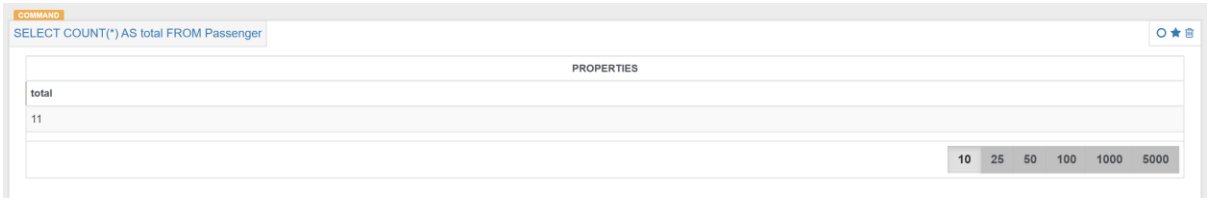
SELECT COUNT(*) AS total FROM Flight

PROPERTIES
total
10

10 25 50 100 1000 5000

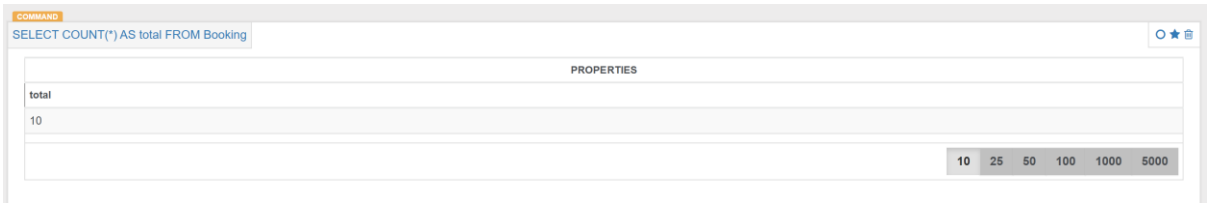
19. Count Passenger Records

SELECT COUNT(*) AS total FROM Passenger



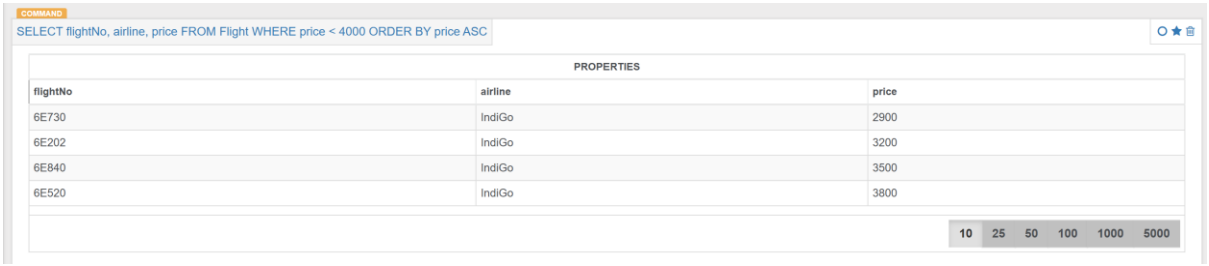
20. Count Booking Records

SELECT COUNT(*) AS total FROM Booking



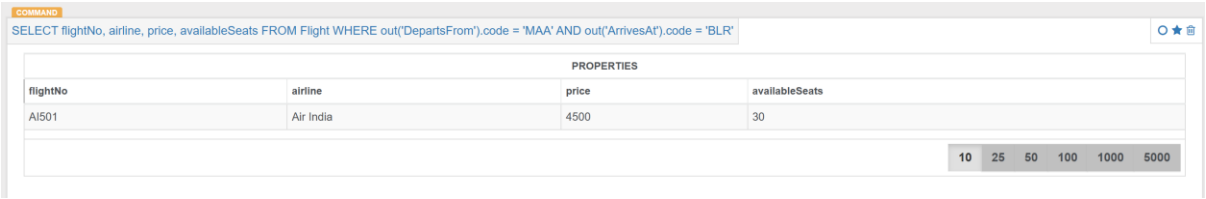
21. Display Flights Below ₹4,000

SELECT flightNo, airline, price FROM Flight WHERE price < 4000 ORDER BY price ASC



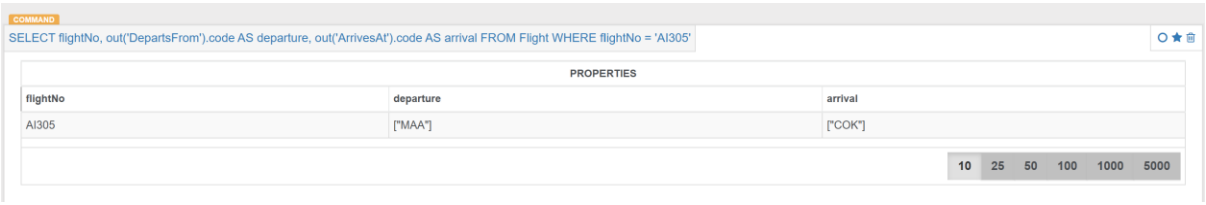
22. Search for a Flight from Chennai to Bengaluru

SELECT flightNo, airline, price, availableSeats FROM Flight WHERE out('DepartsFrom').code = 'MAA' AND out('ArrivesAt').code = 'BLR'



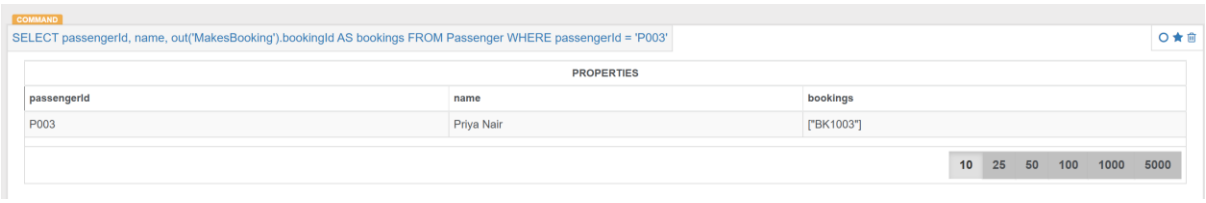
23. Display the Route of Flight AI305

SELECT flightNo, out('DepartsFrom').code AS departure, out('ArrivesAt').code AS arrival FROM Flight WHERE flightNo = 'AI305'



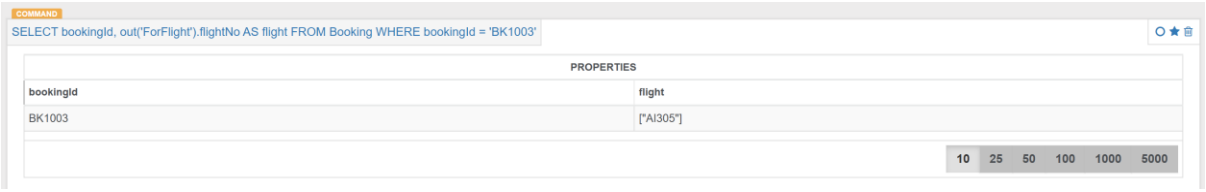
24. Display Passenger P003's Booking

SELECT passengerId, name, out('MakesBooking').bookingId AS bookings FROM Passenger WHERE passengerId = 'P003'



25. Find the Flight for Booking BK1003

SELECT bookingId, out('ForFlight').flightNo AS flight FROM Booking WHERE bookingId = 'BK1003'



26. Display Database Indexes

SELECT expand(indexes) FROM metadata:indexmanager

COMMAND

SELECT expand(indexes) FROM metadata:indexmanager

type	name	indexVersion	indexDefinition	indexDefinitionClass
UNIQUE	Booking.bookingId	4	{"className":"Booking","field":"bookingId","keyType":"STRING","collate":"default","nullValuesIgnored":false}	com.orientechnologies.orient.core.index.OPropertyIn
UNIQUE	OSecurityPolicy.name	4	{"className":"OSecurityPolicy","field":"name","keyType":"STRING","collate":"ci","nullValuesIgnored":false}	com.orientechnologies.orient.core.index.OPropertyIn
UNIQUE	AirportNode.code	4	{"className":"AirportNode","field":"code","keyType":"STRING","collate":"default","nullValuesIgnored":false}	com.orientechnologies.orient.core.index.OPropertyIn
UNIQUE_HASH_INDEX	OFunction.name	3	{"className":"OFunction","field":"name","keyType":"STRING","collate":"default","nullValuesIgnored":false}	com.orientechnologies.orient.core.index.OPropertyIn
UNIQUE	Flight.flightNo	4	{"className":"Flight","field":"flightNo","keyType":"STRING","collate":"default","nullValuesIgnored":false}	com.orientechnologies.orient.core.index.OPropertyIn
UNIQUE	ORole.name	4	{"className":"ORole","field":"name","keyType":"STRING","collate":"ci","nullValuesIgnored":false}	com.orientechnologies.orient.core.index.OPropertyIn
UNIQUE	Passenger.passengerId	4	{"className":"Passenger","field":"passengerId","keyType":"STRING","collate":"default","nullValuesIgnored":false}	com.orientechnologies.orient.core.index.OPropertyIn
UNIQUE	OUser.name	4	{"className":"OUser","field":"name","keyType":"STRING","collate":"ci","nullValuesIgnored":false}	com.orientechnologies.orient.core.index.OPropertyIn

27. Verify Flight AI501

SELECT flightNo, airline, price, availableSeats FROM Flight WHERE flightNo = 'AI501'

COMMAND

SELECT flightNo, airline, price, availableSeats FROM Flight WHERE flightNo = 'AI501'

PROPERTIES

flightNo	airline	price	availableSeats
AI501	Air India	4500	30

10 25 50 100 1000 5000

28. Display Passenger P001

SELECT passengerId, name, email FROM Passenger WHERE passengerId = 'P001'

COMMAND

SELECT passengerId, name, email FROM Passenger WHERE passengerId = 'P001'

PROPERTIES

passengerId	name	email
P001	Ananya Sharma	ananya@example.com

10 25 50 100 1000 5000

29. Display Booking BK1001

SELECT bookingId, seats, totalPrice, status FROM Booking WHERE bookingId = 'BK1001'

COMMAND

SELECT bookingId, seats, totalPrice, status FROM Booking WHERE bookingId = 'BK1001'

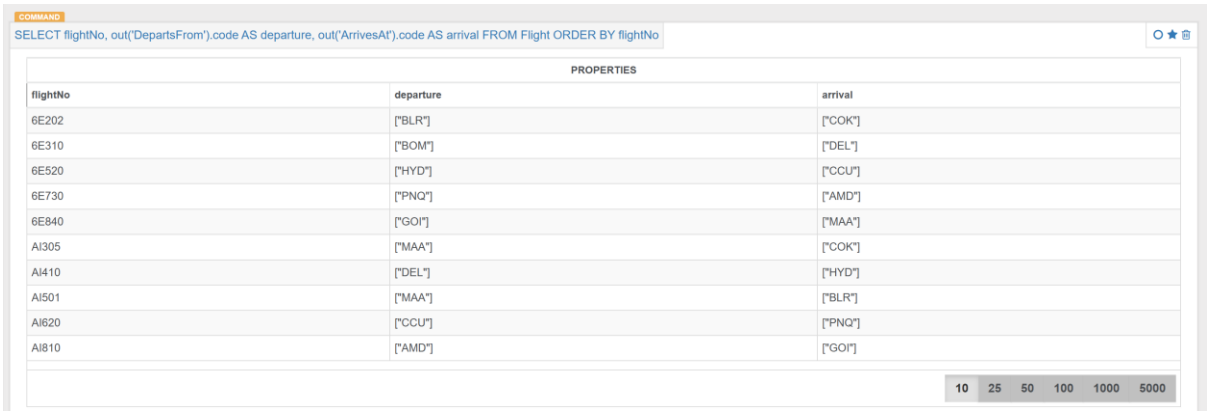
PROPERTIES

bookingId	seats	totalPrice	status
BK1001	2	9000	CONFIRMED

10 25 50 100 1000 5000

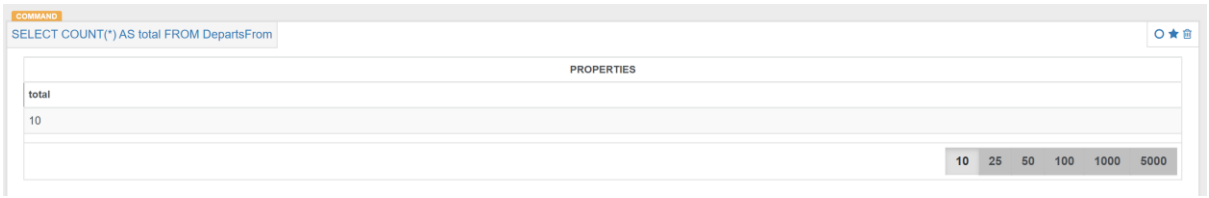
30. Display All Flights in Route Format

SELECT flightNo, out('DepartsFrom').code AS departure, out('ArrivesAt').code AS arrival FROM Flight ORDER BY flightNo



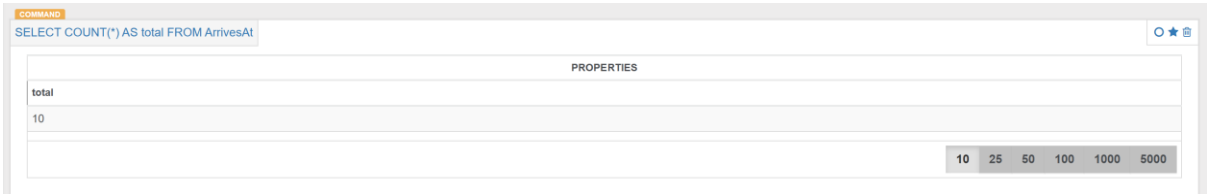
31. Count Departure Relationships

SELECT COUNT(*) AS total FROM DepartsFrom



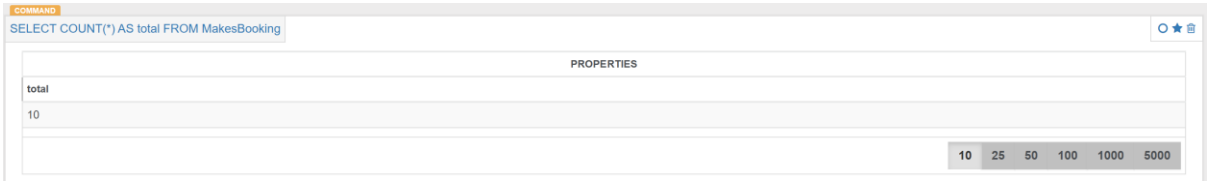
32. Count Arrival Relationships

SELECT COUNT(*) AS total FROM ArrivesAt



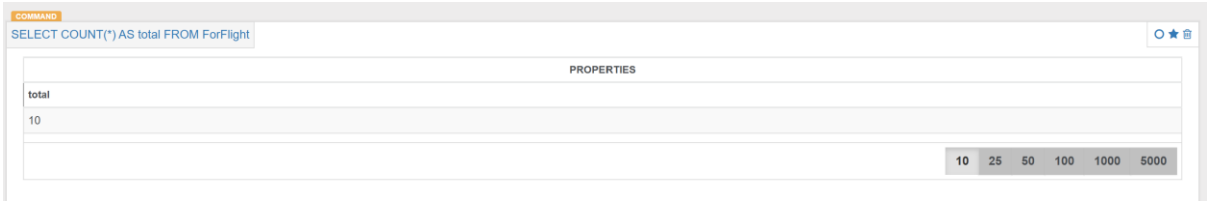
33. Count Passenger–Booking Relationships

SELECT COUNT(*) AS total FROM MakesBooking



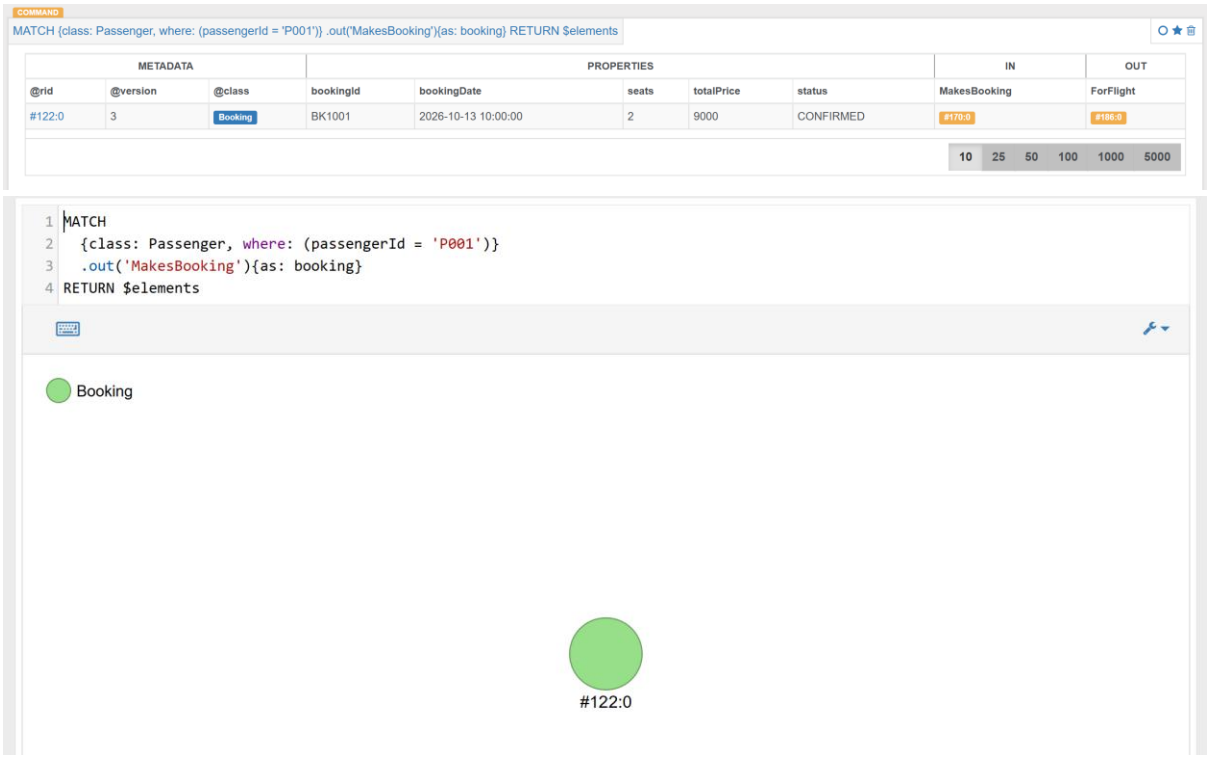
34. Count Booking–Flight Relationships

SELECT COUNT(*) AS total FROM ForFlight



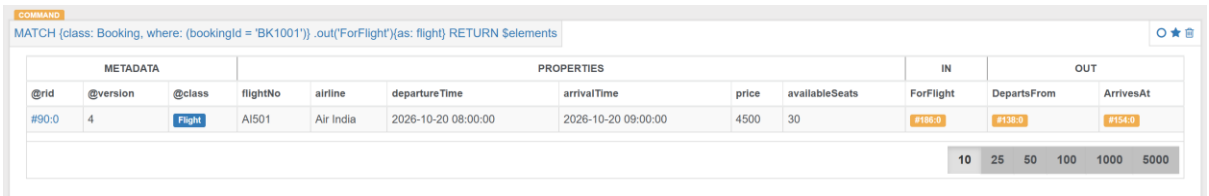
35. Passenger to Booking Graph

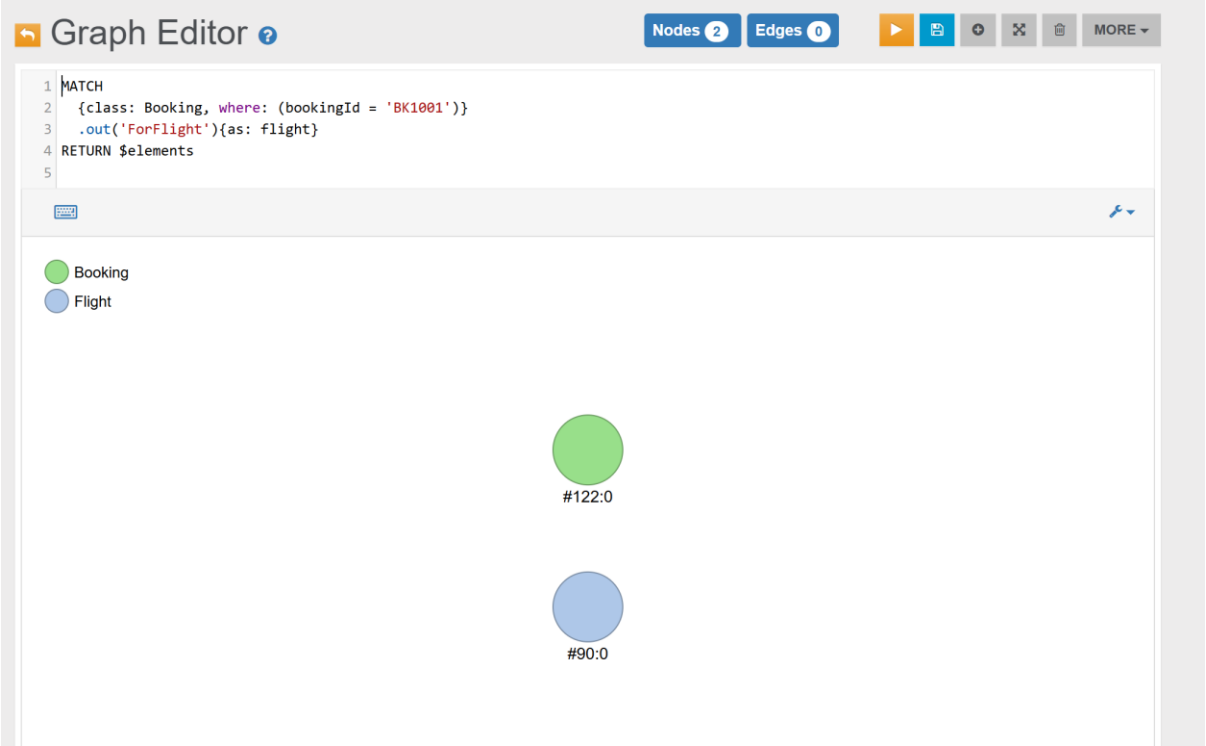
```
MATCH
 {class: Passenger, where: (passengerId = 'P001')}
 .out('MakesBooking'){as: booking}
RETURN $elements
```



36. Booking to Flight Graph

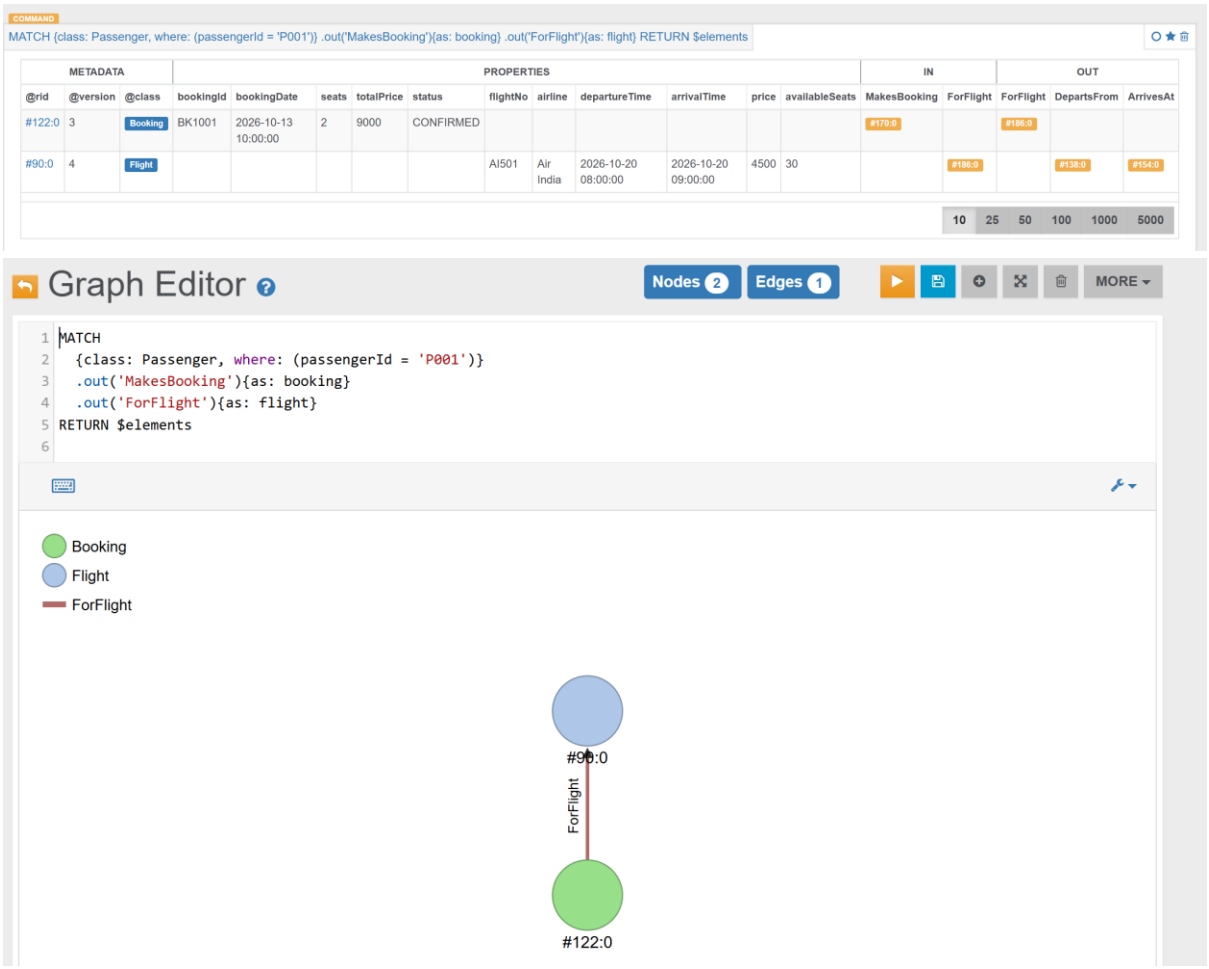
```
MATCH
 {class: Booking, where: (bookingId = 'BK1001')}
 .out('ForFlight'){as: flight}
RETURN $elements
```





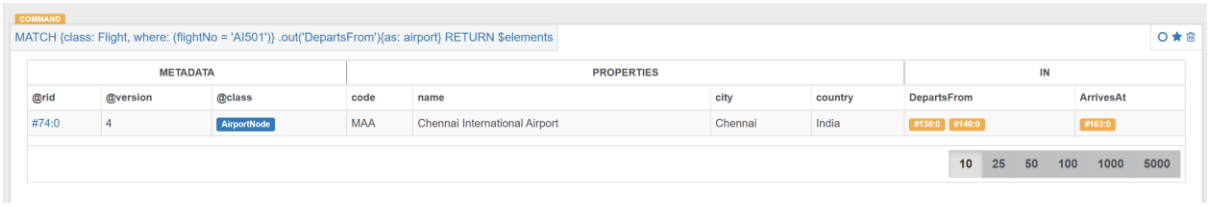
37. Complete Passenger–Booking–Flight Graph

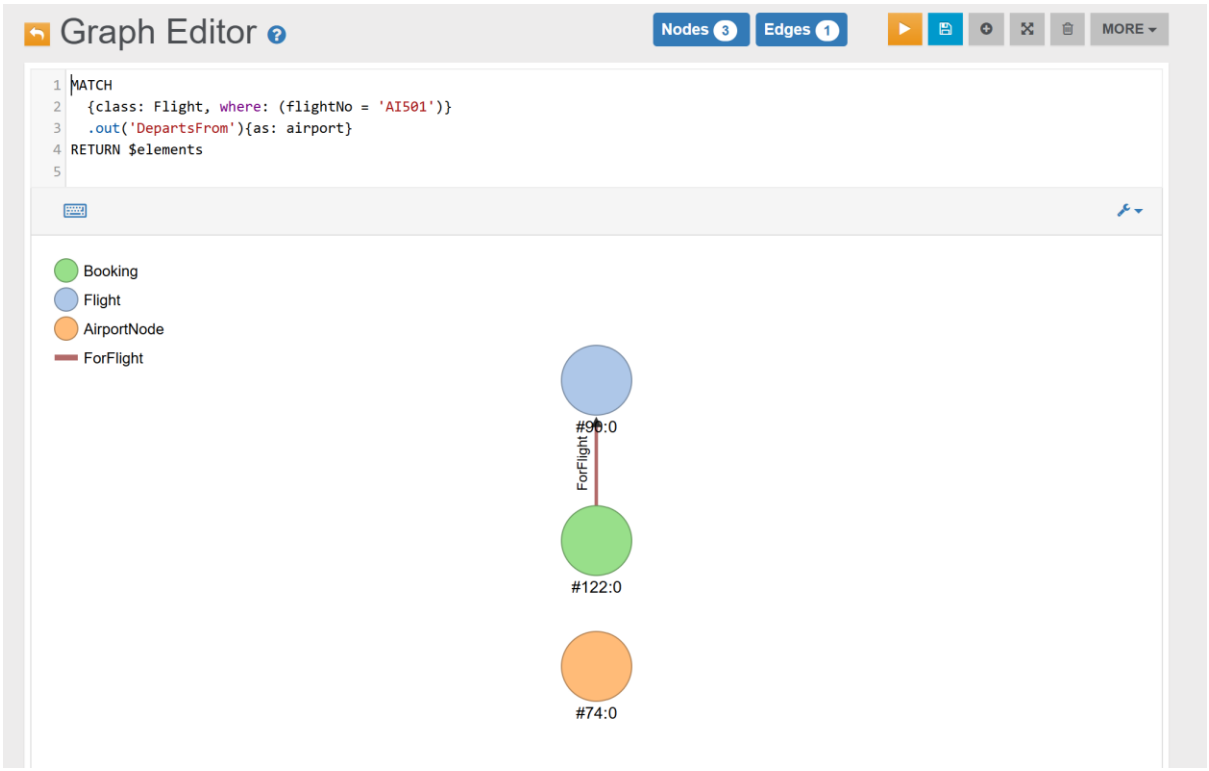
```
MATCH
 {class: Passenger, where: (passengerId = 'P001')}
 .out('MakesBooking'){as: booking}
 .out('ForFlight'){as: flight}
RETURN $elements
```



38. Flight Departure Airport Graph

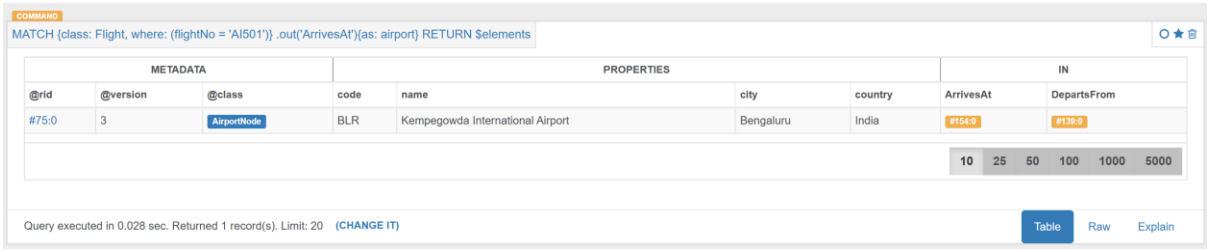
```
MATCH
 {class: Flight, where: (flightNo = 'AI501')}
 .out('DepartsFrom'){as: airport}
RETURN $elements
```

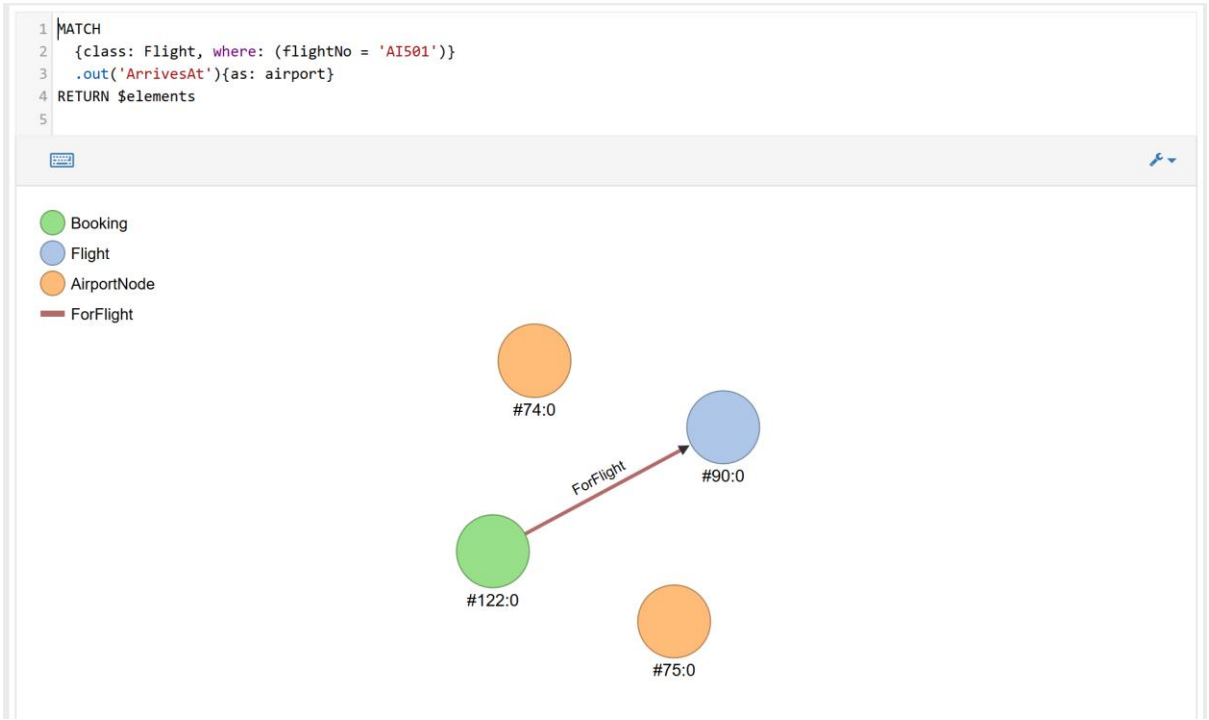



Graph Editor
Nodes 3
Edges 1
MORE
MATCH
{class: Flight, where: (flightNo = 'AI501')}
.out('DepartsFrom'){as: airport}
RETURN $elements
Booking
Flight
AirportNode
ForFlight
#90:0
ForFlight
#122:0
#74:0

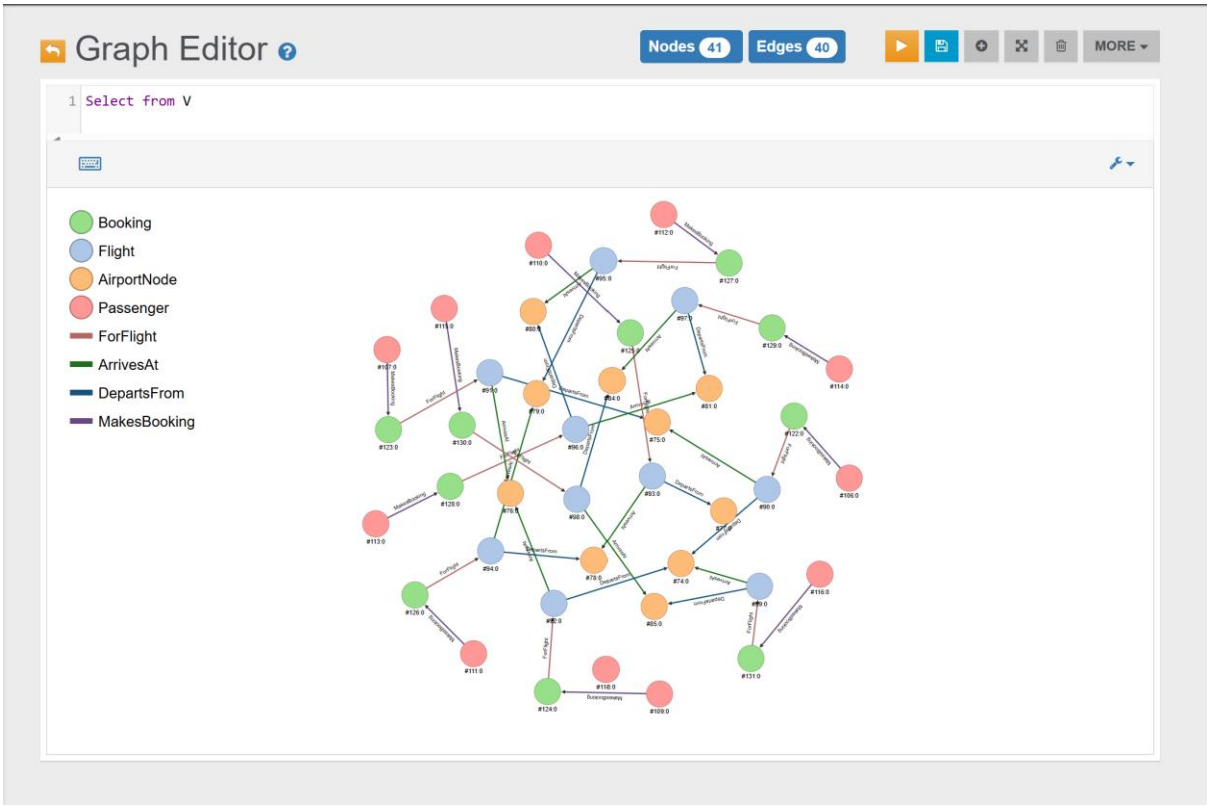
39. Flight Arrival Airport Graph

```
MATCH
 {class: Flight, where: (flightNo = 'AI501')}
 .out('ArrivesAt'){as: airport}
RETURN $elements
```

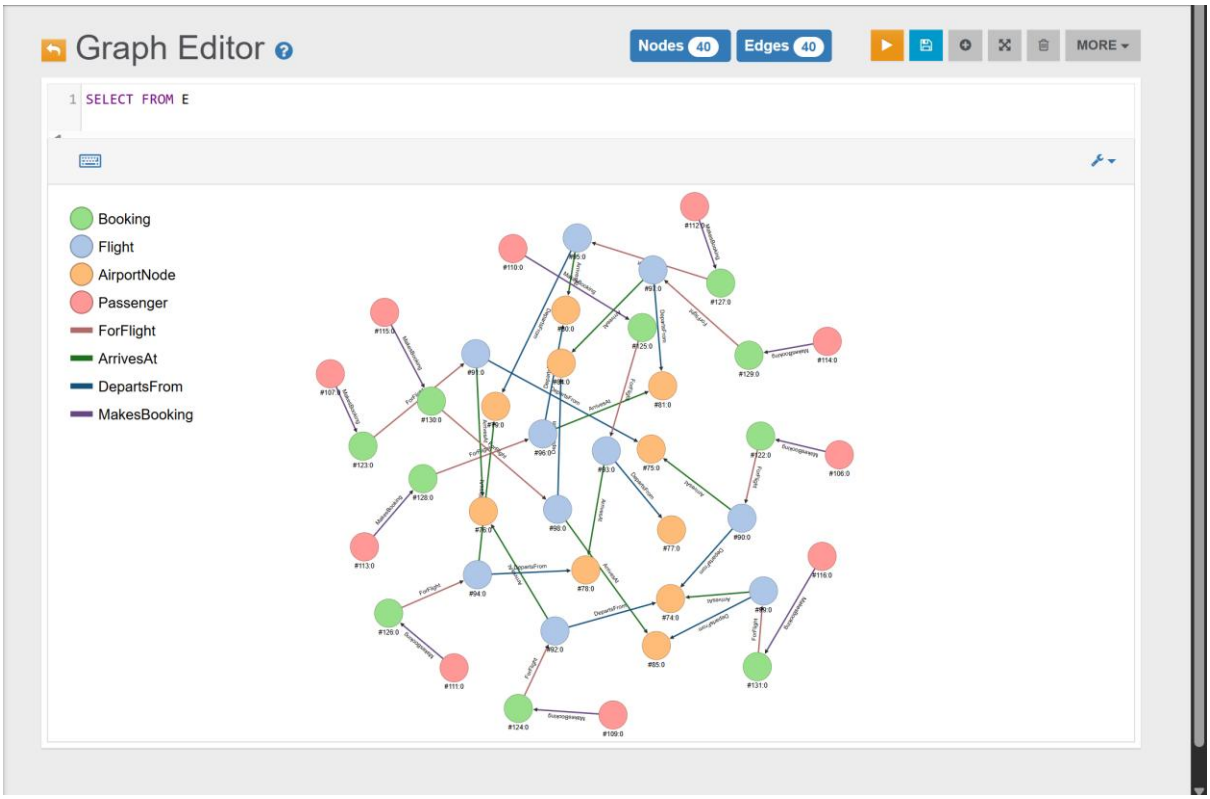




40. Vertex Class



41. Edge Class



Initial Queries

1. Installation using Docker

```
docker run -d --name orientdb \
 -p 2424:2424 -p 2480:2480 \
 -e ORIENTDB_ROOT_PASSWORD=root123 \
 orientdb:3.2
```

Studio URL: http://localhost:2480

2. Connect to the server

```
CONNECT remote:localhost root root123
```

3. Create the database

```
CREATE DATABASE remote:localhost/FlightDB root root123 plocal
```

4. Connect to FlightDB

```
CONNECT remote:localhost/FlightDB admin admin
```

Use the credentials configured for your server. The example credentials above may not match yours.

5. Create AirportNode class, properties and index

```
CREATE CLASS AirportNode EXTENDS V;
CREATE PROPERTY AirportNode.code STRING;
CREATE PROPERTY AirportNode.name STRING;
CREATE INDEX AirportNode.code ON AirportNode(code) UNIQUE;
```

6. Create Flight class and index

```
CREATE CLASS Flight EXTENDS V;
CREATE PROPERTY Flight.flight_no STRING;
CREATE INDEX Flight.flight_no ON Flight(flight_no) UNIQUE;
```

7. Create Passenger class

```
CREATE CLASS Passenger EXTENDS V;
```

8. Create Booking class

```
CREATE CLASS Booking EXTENDS V;
```

9. Create edge classes

```
CREATE CLASS ForFlight EXTENDS E;
CREATE CLASS ArrivesAt EXTENDS E;
CREATE CLASS DepartsFrom EXTENDS E;
```

```
CREATE CLASS MakesBooking EXTENDS E;
```

10. Insert airport vertices

```
CREATE VERTEX AirportNode
SET code = 'MAA',
  name = 'Chennai Intl',
  city = 'Chennai';

CREATE VERTEX AirportNode
SET code = 'DEL',
  name = 'Delhi Intl',
  city = 'Delhi';
```

11. Insert a flight vertex

```
CREATE VERTEX Flight
SET flight_no = 'AI101',
  seats = 180,
  fare = 5200;
```

12. Connect Chennai to flight AI101

```
CREATE EDGE DepartsFrom
FROM (SELECT FROM AirportNode WHERE code = 'MAA')
TO (SELECT FROM Flight WHERE flight_no = 'AI101');
```

13. Connect flight AI101 to Delhi

```
CREATE EDGE ArrivesAt
FROM (SELECT FROM Flight WHERE flight_no = 'AI101')
TO (SELECT FROM AirportNode WHERE code = 'DEL');
```

14. Create a passenger

```
CREATE VERTEX Passenger
SET pid = 1,
  name = 'Asha',
  email = 'asha@x.com';
```

15. Retrieve flights with fares below 6000

```
SELECT FROM Flight
WHERE fare < 6000
ORDER BY fare;
```

16. Update a flight's fare

```
UPDATE Flight
SET fare = 5500
WHERE flight_no = 'AI101';
```

17. Find destinations from Chennai

```
SELECT expand(out('DepartsFrom').out('ArrivesAt'))
FROM AirportNode
WHERE code = 'MAA';
```

18. Find all bookings made by Asha

```
SELECT expand(out('MakesBooking'))
FROM Passenger
WHERE name = 'Asha';
```

19. Find flights booked by Asha and their destination cities

```
MATCH
 {class: Passenger, as: p,
  where: (name = 'Asha')}
 .out('MakesBooking'){as: b}
 .out('ForFlight'){as: f}
 .out('ArrivesAt'){as: a}

RETURN
 p.name AS passenger,
 f.flight_no AS flight,
 a.city AS city;
```

20. Find the shortest path from Chennai to Delhi

```
SELECT expand(shortestPath($a, $b, 'OUT'))
LET
 $a = (SELECT FROM AirportNode WHERE code = 'MAA'),
 $b = (SELECT FROM AirportNode WHERE code = 'DEL');
```

21. Book a flight transactionally

```
BEGIN;
UPDATE Flight
SET seats = seats - 1
WHERE flight_no = 'AI101';
CREATE EDGE MakesBooking
FROM (SELECT FROM Passenger WHERE pid = 1)
TO (CREATE VERTEX Booking SET seat_no = '12A');
CREATE EDGE ForFlight
FROM (SELECT FROM Booking WHERE seat_no = '12A')
TO (SELECT FROM Flight WHERE flight_no = 'AI101');
COMMIT;
```

22. Create an index on the booking seat number

```
CREATE INDEX Booking.seat
ON Booking(seat_no) NOTUNIQUE;
```